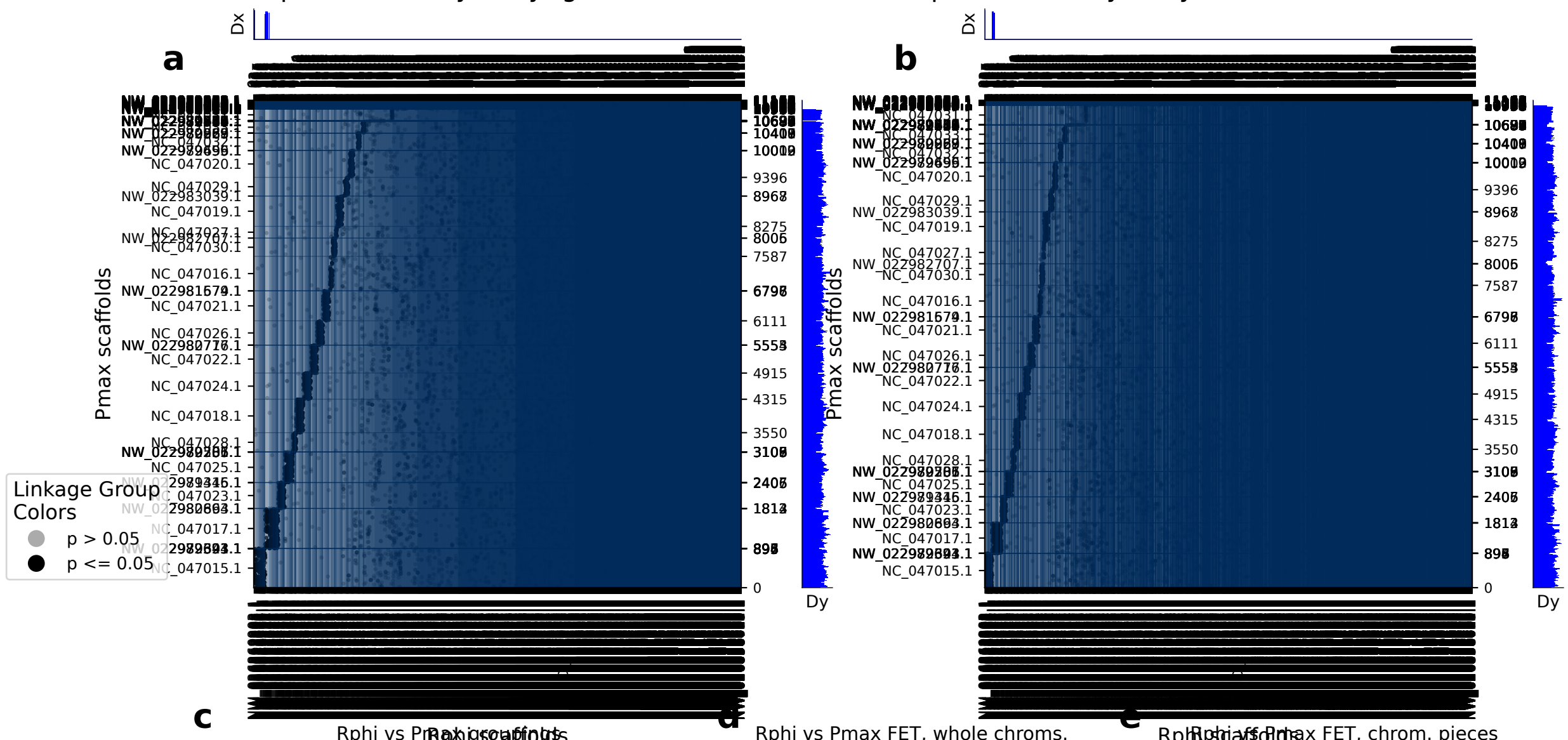


**diamond: 11163 orthologs on 4400 Rphi scaffolds and 246 Pmax scaffolds.**



## Rphi vs Pn

## Rphi vs Pmax FET, whole chroms.

RpH<sub>1</sub> scaffold, max FET, chrom. pieces

|    | Rphi scaf      | Pmax scaf   | p (FET)  | Court |
|----|----------------|-------------|----------|-------|
| 0  | NW_026860903.1 | NC_047015.1 | 4.05e-21 | 24    |
| 1  | NW_026865571.1 | NC_047017.1 | 2.83e-15 | 21    |
| 2  | NW_026861563.1 | NC_047017.1 | 2.09e-14 | 21    |
| 3  | NW_026865484.1 | NC_047017.1 | 3.23e-14 | 20    |
| 4  | NW_026861019.1 | NC_047017.1 | 2.47e-12 | 19    |
| 5  | NW_026852258.1 | NC_047023.1 | 4.64e-19 | 19    |
| 6  | NW_026862663.1 | NC_047015.1 | 1.99e-10 | 17    |
| 7  | NW_026859933.1 | NC_047018.1 | 2.50e-13 | 17    |
| 8  | NW_026852008.1 | NC_047015.1 | 2.20e-13 | 17    |
| 9  | NW_026862783.1 | NC_047015.1 | 5.23e-10 | 15    |
| 10 | NW_026855294.1 | NC_047025.1 | 2.51e-09 | 15    |
| 11 | NW_026853802.1 | NC_047015.1 | 3.53e-11 | 15    |
| 12 | NW_026851633.1 | NC_047026.1 | 3.97e-13 | 15    |
| 13 | NW_026865967.1 | NC_047018.1 | 6.86e-10 | 14    |
| 14 | NW_026858503.1 | NC_047021.1 | 1.45e-10 | 14    |
| 15 | NW_026852000.1 | NC_047016.1 | 7.55e-11 | 14    |
| 16 | NW_026867184.1 | NC_047030.1 | 3.47e-12 | 13    |
| 17 | NW_026863645.1 | NC_047028.1 | 1.14e-08 | 13    |
| 18 | NW_026863541.1 | NC_047017.1 | 1.44e-09 | 13    |
| 19 | NW_026859561.1 | NC_047024.1 | 2.99e-11 | 13    |
| 20 | NW_026854960.1 | NC_047018.1 | 9.49e-09 | 13    |
| 21 | NW_026852705.1 | NC_047015.1 | 2.52e-06 | 13    |
| 22 | NW_026867313.1 | NC_047021.1 | 2.82e-09 | 12    |
| 23 | NW_026863896.1 | NC_047025.1 | 2.86e-07 | 12    |
| 24 | NW_026853170.1 | NC_047015.1 | 8.60e-07 | 12    |
| 25 | NW_026852468.1 | NC_047017.1 | 7.16e-06 | 12    |
| 26 | NW_026864352.1 | NC_047018.1 | 1.78e-06 | 11    |
| 27 | NW_026862474.1 | NC_047019.1 | 3.63e-06 | 11    |
| 28 | NW_026861864.1 | NC_047025.1 | 1.88e-04 | 11    |
| 29 | NW_026861840.1 | NC_047024.1 | 9.86e-05 | 11    |
| 30 | NW_026861574.1 | NC_047019.1 | 5.22e-08 | 11    |
| 31 | NW_026859998.1 | NC_047018.1 | 1.58e-07 | 11    |
| 32 | NW_026859790.1 | NC_047017.1 | 1.26e-05 | 11    |
| 33 | NW_026859418.1 | NC_047023.1 | 9.07e-09 | 11    |
| 34 | NW_026859313.1 | NC_047019.1 | 5.22e-08 | 11    |
| 35 | NW_026852998.1 | NC_047019.1 | 5.22e-08 | 11    |
| 36 | NW_026866838.1 | NC_047026.1 | 9.41e-08 | 10    |
| 37 | NW_026866837.1 | NC_047017.1 | 1.43e-04 | 10    |
| 38 | NW_026866672.1 | NC_047024.1 | 2.03e-07 | 10    |
| 39 | NW_026866080.1 | NC_047017.1 | 1.40e-05 | 10    |
| 40 | NW_026865813.1 | NC_047023.1 | 1.04e-05 | 10    |
| 41 | NW_026865227.1 | NC_047022.1 | 3.76e-07 | 10    |
| 42 | NW_026863691.1 | NC_047021.1 | 8.01e-06 | 10    |
| 43 | NW_026862689.1 | NC_047017.1 | 3.18e-03 | 10    |
| 44 | NW_026861084.1 | NC_047025.1 | 2.24e-04 | 10    |
| 45 | NW_026860451.1 | NC_047018.1 | 1.36e-04 | 10    |
| 46 | NW_026859959.1 | NC_047029.1 | 6.71e-09 | 10    |
| 47 | NW_026859620.1 | NC_047017.1 | 7.93e-04 | 10    |
| 48 | NW_026859162.1 | NC_047026.1 | 9.41e-08 | 10    |
| 49 | NW_026859085.1 | NC_047015.1 | 1.13e-05 | 10    |
| 50 | NW_026857770.1 | NC_047025.1 | 4.75e-05 | 10    |
| 51 | NW_026856137.1 | NC_047017.1 | 1.40e-05 | 10    |
| 52 | NW_026855828.1 | NC_047018.1 | 2.42e-05 | 10    |
| 53 | NW_026855648.1 | NC_047015.1 | 1.16e-04 | 10    |
| 54 | NW_026855197.1 | NC_047022.1 | 5.07e-03 | 10    |
| 55 | NW_026853905.1 | NC_047017.1 | 1.40e-05 | 10    |
| 56 | NW_026852806.1 | NC_047030.1 | 5.28e-09 | 10    |
| 57 | NW_026852123.1 | NC_047021.1 | 7.70e-07 | 10    |
| 58 | NW_026851646.1 | NC_047026.1 | 9.41e-08 | 10    |
| 59 | NW_026867092.1 | NC_047015.1 | 1.33e-03 | 9     |
| 60 | NW_026866929.1 | NC_047023.1 | 3.66e-04 | 9     |
| 61 | NW_026866451.1 | NC_047018.1 | 3.24e-04 | 9     |
| 62 | NW_026866098.1 | NC_047015.1 | 1.33e-03 | 9     |
| 63 | NW_026865464.1 | NC_047026.1 | 3.69e-04 | 9     |

|    | Rphi scaf              | Pmax scaf           | p (FET)    | Count |
|----|------------------------|---------------------|------------|-------|
| 0  | NW_026860903.1 : 0-end | NC_047015.1 : 0-end | 4.05e-2124 |       |
| 1  | NW_026865571.1 : 0-end | NC_047017.1 : 0-end | 2.83e-1521 |       |
| 2  | NW_026861563.1 : 0-end | NC_047017.1 : 0-end | 2.09e-1421 |       |
| 3  | NW_026865484.1 : 0-end | NC_047017.1 : 0-end | 3.23e-1420 |       |
| 4  | NW_026861019.1 : 0-end | NC_047017.1 : 0-end | 2.47e-1219 |       |
| 5  | NW_026852258.1 : 0-end | NC_047023.1 : 0-end | 4.64e-1919 |       |
| 6  | NW_026862663.1 : 0-end | NC_047015.1 : 0-end | 1.99e-1017 |       |
| 7  | NW_026859933.1 : 0-end | NC_047018.1 : 0-end | 2.50e-1317 |       |
| 8  | NW_026852008.1 : 0-end | NC_047015.1 : 0-end | 2.20e-1317 |       |
| 9  | NW_026862783.1 : 0-end | NC_047015.1 : 0-end | 5.23e-1015 |       |
| 10 | NW_026855294.1 : 0-end | NC_047025.1 : 0-end | 2.51e-0915 |       |
| 11 | NW_026853802.1 : 0-end | NC_047015.1 : 0-end | 3.53e-1115 |       |
| 12 | NW_026851633.1 : 0-end | NC_047026.1 : 0-end | 3.97e-1315 |       |
| 13 | NW_026865967.1 : 0-end | NC_047018.1 : 0-end | 6.86e-1014 |       |
| 14 | NW_026858503.1 : 0-end | NC_047021.1 : 0-end | 1.45e-1014 |       |
| 15 | NW_026852000.1 : 0-end | NC_047016.1 : 0-end | 7.55e-1114 |       |
| 16 | NW_026867184.1 : 0-end | NC_047030.1 : 0-end | 3.47e-1213 |       |
| 17 | NW_026863645.1 : 0-end | NC_047028.1 : 0-end | 1.14e-0813 |       |
| 18 | NW_026863541.1 : 0-end | NC_047017.1 : 0-end | 7.44e-0913 |       |
| 19 | NW_026859561.1 : 0-end | NC_047024.1 : 0-end | 2.99e-1113 |       |
| 20 | NW_026854960.1 : 0-end | NC_047018.1 : 0-end | 9.49e-0913 |       |
| 21 | NW_026852705.1 : 0-end | NC_047015.1 : 0-end | 2.52e-0613 |       |
| 22 | NW_026867313.1 : 0-end | NC_047021.1 : 0-end | 2.82e-0912 |       |
| 23 | NW_026863896.1 : 0-end | NC_047025.1 : 0-end | 2.86e-0712 |       |
| 24 | NW_026853170.1 : 0-end | NC_047015.1 : 0-end | 8.60e-0712 |       |
| 25 | NW_026852468.1 : 0-end | NC_047017.1 : 0-end | 7.16e-0612 |       |
| 26 | NW_026864352.1 : 0-end | NC_047018.1 : 0-end | 1.78e-0611 |       |
| 27 | NW_026862474.1 : 0-end | NC_047019.1 : 0-end | 3.63e-0611 |       |
| 28 | NW_026861864.1 : 0-end | NC_047025.1 : 0-end | 1.88e-0411 |       |
| 29 | NW_026861840.1 : 0-end | NC_047024.1 : 0-end | 9.86e-0511 |       |
| 30 | NW_026861574.1 : 0-end | NC_047019.1 : 0-end | 5.22e-0811 |       |
| 31 | NW_026859998.1 : 0-end | NC_047018.1 : 0-end | 1.58e-0711 |       |
| 32 | NW_026859790.1 : 0-end | NC_047017.1 : 0-end | 1.26e-0511 |       |
| 33 | NW_026859418.1 : 0-end | NC_047023.1 : 0-end | 9.07e-0911 |       |
| 34 | NW_026859313.1 : 0-end | NC_047019.1 : 0-end | 5.22e-0811 |       |
| 35 | NW_026852998.1 : 0-end | NC_047019.1 : 0-end | 5.22e-0811 |       |
| 36 | NW_026866838.1 : 0-end | NC_047026.1 : 0-end | 9.41e-0810 |       |
| 37 | NW_026866837.1 : 0-end | NC_047017.1 : 0-end | 1.43e-0410 |       |
| 38 | NW_026866672.1 : 0-end | NC_047024.1 : 0-end | 2.03e-0710 |       |
| 39 | NW_026866080.1 : 0-end | NC_047017.1 : 0-end | 1.40e-0510 |       |
| 40 | NW_026865813.1 : 0-end | NC_047023.1 : 0-end | 1.04e-0510 |       |
| 41 | NW_026865227.1 : 0-end | NC_047022.1 : 0-end | 3.76e-0710 |       |
| 42 | NW_026863691.1 : 0-end | NC_047021.1 : 0-end | 8.01e-0610 |       |
| 43 | NW_026862689.1 : 0-end | NC_047017.1 : 0-end | 3.18e-0310 |       |
| 44 | NW_026861084.1 : 0-end | NC_047025.1 : 0-end | 2.24e-0410 |       |
| 45 | NW_026860451.1 : 0-end | NC_047018.1 : 0-end | 1.36e-0410 |       |
| 46 | NW_026859959.1 : 0-end | NC_047029.1 : 0-end | 6.71e-0910 |       |
| 47 | NW_026859620.1 : 0-end | NC_047017.1 : 0-end | 7.93e-0410 |       |
| 48 | NW_026859162.1 : 0-end | NC_047026.1 : 0-end | 9.41e-0810 |       |
| 49 | NW_026859085.1 : 0-end | NC_047015.1 : 0-end | 1.13e-0510 |       |
| 50 | NW_026857770.1 : 0-end | NC_047025.1 : 0-end | 5.47e-0510 |       |
| 51 | NW_026856137.1 : 0-end | NC_047017.1 : 0-end | 1.40e-0510 |       |
| 52 | NW_026855828.1 : 0-end | NC_047018.1 : 0-end | 2.42e-0510 |       |
| 53 | NW_026855648.1 : 0-end | NC_047015.1 : 0-end | 1.16e-0410 |       |
| 54 | NW_026855197.1 : 0-end | NC_047022.1 : 0-end | 5.07e-0310 |       |
| 55 | NW_026853905.1 : 0-end | NC_047017.1 : 0-end | 1.40e-0510 |       |
| 56 | NW_026852806.1 : 0-end | NC_047030.1 : 0-end | 5.28e-0910 |       |
| 57 | NW_026852123.1 : 0-end | NC_047021.1 : 0-end | 7.70e-0710 |       |
| 58 | NW_026851646.1 : 0-end | NC_047026.1 : 0-end | 9.41e-0810 |       |
| 59 | NW_026867092.1 : 0-end | NC_047015.1 : 0-end | 1.33e-0310 |       |
| 60 | NW_026866929.1 : 0-end | NC_047023.1 : 0-end | 6.36e-049  |       |
| 61 | NW_026866451.1 : 0-end | NC_047018.1 : 0-end | 3.24e-049  |       |
| 62 | NW_026866098.1 : 0-end | NC_047015.1 : 0-end | 1.33e-039  |       |
| 63 | NW_026865464.1 : 0-end | NC_047026.1 : 0-end | 3.69e-049  |       |

(a) depicts the chromosomes/scaffolds of Rphi (x-axis) plotted against the chromosomes/scaffolds of Pmax (y-axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 11163 orthologs found between 4400 Rphi scaffolds and 246 Pmax scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chromosome FET p-value , and is not a FET value of the correlation between the chromosome pair and the gene group. (c) shows chromosome-scale significance values. This table shows the same information as c, but does not show a gene group information. (d) shows the same information as c, but does not show a gene group information. Useful for showing if single arms of chromosomes are correlated with other regions.

|      | NW_026866130.1 | NC_047025.1 | FET p-value | Dx | Dy |
|------|----------------|-------------|-------------|----|----|
| 1    | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 2    | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 3    | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 4    | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 5    | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 6    | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 7    | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 8    | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 9    | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 10   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 11   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 12   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 13   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 14   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 15   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 16   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 17   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 18   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 19   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 20   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 21   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 22   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 23   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 24   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 25   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 26   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 27   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 28   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 29   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 30   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 31   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 32   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 33   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 34   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 35   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 36   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 37   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 38   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 39   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 40   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 41   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 42   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 43   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 44   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 45   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 46   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 47   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 48   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 49   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 50   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 51   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 52   | NW_026866130.1 | NC_047025.1 | 2.79e-03    | 0  | 0  |
| 53</ |                |             |             |    |    |